

AI-Assisted Emergency Department Triage

Evaluating the Operational Impact of AI-Assisted Triage Using a Randomized A/B Testing Framework

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Tools: Python | Pandas | NumPy | SciPy | Plotly | Power BI | DAX

7.4 min faster Door-to-Doctor Time	15.8% improvement vs. standard triage	$p = 8.8 \times 10^{-17}$ statistically significant result
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Project Snapshot

Industry	Healthcare
Project Type	Randomized A/B Testing
Hospital	Riverside General Hospital (Simulated)
Study Period	January–March 2026
Primary Objective	Evaluate whether AI-assisted triage reduces Door-to-Doctor Time
Primary KPI	Door-to-Doctor Time
Sample Size	2,253 Emergency Department encounters
Control Group	1,125 patients
Treatment Group	1,128 patients
Tools Used	Python, SciPy, Pandas, Plotly, Power BI, DAX
Outcome	AI-assisted triage reduced Door-to-Doctor Time by 7.4 minutes (15.8%) while increasing Door-to-Triage Time by only 7 seconds.

Executive Summary

Emergency Departments (EDs) are among the busiest and most resource-intensive units within a hospital. One of the most closely monitored operational metrics is Door-to-Doctor Time, the interval between a patient's arrival and their first assessment by a physician. Delays in this process contribute to overcrowding, prolonged waiting times, reduced patient satisfaction, and delayed clinical intervention.

This project evaluates the operational impact of an AI-assisted triage workflow using a randomized A/B testing framework. A simulated dataset representing 2,253 Emergency Department encounters from Riverside General Hospital was used to compare a traditional nurse-led triage process (Control group) with an AI-assisted triage workflow (Treatment group). Randomization ensured that both groups were comparable, allowing differences in performance to be attributed to the intervention rather than underlying patient characteristics.

The analysis demonstrated that the AI-assisted workflow reduced the average Door-to-Doctor Time by 7.4 minutes, a 15.8% improvement over the standard triage process. Statistical testing produced a highly significant result ($p = 8.828 \times 10^{-17}$), while the 95% confidence interval and Cohen's d indicated that the improvement was both reliable and operationally meaningful. Importantly, the intervention increased Door-to-Triage Time by only about seven seconds, suggesting the efficiency gained later in the patient journey came at virtually no cost to the triage process itself.

Based on these results, a phased pilot implementation is recommended to validate the observed benefits within a live clinical environment before broader deployment.

1. Business Background & Problem

Emergency Departments serve as the primary point of care for patients experiencing acute illnesses and emergencies. Unlike most hospital departments, EDs operate continuously and must respond to unpredictable fluctuations in patient volume, clinical complexity, and resource availability. One key contributor to ED congestion is delayed physician assessment; patients who wait longer occupy valuable clinical space, contribute to overcrowding, and increase pressure on healthcare professionals.

Riverside General Hospital had experienced increasing operational pressure within its Emergency Department due to sustained growth in patient demand. Leadership observed that patients were waiting longer than expected for their initial physician assessment, contributing to congestion, reduced throughput, and increasing strain on clinical staff.

An AI-assisted triage solution had been proposed as a potential intervention, but leadership recognized that implementing it would require financial investment, workflow redesign, staff training, and ongoing governance. Adopting the technology without objective evidence of its effectiveness could introduce unnecessary operational risk without delivering meaningful improvement. The central challenge, therefore, was not simply whether an AI-assisted triage system could be implemented, but whether it could demonstrably improve ED performance, evidence that improvement resulted from the intervention itself rather than random variation or differences in patient characteristics.

Business Questions

1. Does an AI-assisted triage workflow reduce Door-to-Doctor Time compared with the current nurse-led process?
2. Does introducing AI increase Door-to-Triage Time and create additional delays during patient assessment?
3. Were patients assigned to the Control and Treatment groups sufficiently comparable to ensure a fair evaluation?
4. Are any observed differences statistically significant and practically meaningful from an operational perspective?
5. Based on the available evidence, should Riverside General Hospital proceed with implementing the AI-assisted triage system?

Project Objectives

- Evaluate the operational impact of AI-assisted triage on physician waiting times.
- Validate the effectiveness of the randomization process by comparing baseline characteristics between groups.
- Determine whether observed improvements were statistically significant using appropriate hypothesis testing techniques.
- Assess the practical significance of the intervention through confidence intervals and effect size analysis.
- Translate statistical findings into actionable business recommendations for evidence-based decision-making.

2. Methodology

Experiment Design

To ensure that any observed improvement could be confidently attributed to the AI-assisted triage system, this study adopted a randomized A/B testing framework. The existing nurse-led triage workflow served as the Control group, while the AI-assisted triage workflow represented the Treatment group. Patients arriving at the Emergency Department were randomly assigned to one of the two workflows, giving every patient an equal probability of receiving either intervention.

Random assignment minimizes selection bias and helps produce groups with comparable characteristics. When treatment and control populations are similar at baseline, differences observed afterward are more likely to result from the intervention itself. The experimental unit was the individual Emergency Department encounter, including operational variables such as Door-to-Triage Time, Door-to-Doctor Time, patient demographics, triage level, and presenting complaint.

The experiment used a simulated dataset of 2,253 Emergency Department encounters, with 1,125 patients assigned to the Control group and 1,128 to the AI-assisted Treatment group, a near-equal allocation that ensured a balanced comparison while maintaining adequate statistical power.

Dataset Overview

The analysis used a synthetic dataset representing Emergency Department operations at Riverside General Hospital over three months (January–March 2026), constructed to reflect realistic patient characteristics, operational workflows, and performance metrics.

- Age — included to confirm comparable age distributions between groups, since age often influences disease severity and resource utilization.
- Gender — used to assess whether randomization produced comparable populations across both groups.
- Chief Complaint — each patient's primary reason for visiting, providing operational context for prioritization and physician workload.
- Triage Level — the clinical urgency assigned during assessment, checked for balance between groups.
- Door-to-Triage Time — interval between arrival and completed triage assessment.
- Door-to-Doctor Time — interval between arrival and first physician assessment (the study's primary outcome).

Success Metrics

Door-to-Doctor Time was selected as the primary performance metric, directly reflecting one of the most important indicators of Emergency Department efficiency. Door-to-Triage Time was tracked as a secondary metric to confirm that introducing AI did not unintentionally delay earlier stages of the patient journey. Performance Improvement (%) and Patient Distribution across groups were tracked as supporting indicators of both efficiency and the fairness of the randomization.

Exploratory Data Analysis

Before hypothesis testing, an exploratory analysis validated that randomization had produced comparable groups. The split of 1,125 Control vs. 1,128 Treatment patients showed an almost perfectly balanced allocation. Age, gender, and triage-level distributions were comparable across groups, and chief complaints, abdominal pain, fever, chest pain, trauma, and respiratory complaints appeared across both groups, reflecting a realistic ED population rather than a narrow one.

Initial visualizations suggested that patients in the AI-assisted workflow generally experienced shorter Door-to-Doctor Times, while Door-to-Triage Times appeared nearly identical between groups, early evidence that the dataset was suitable for formal statistical testing.

3. Statistical Analysis

An Independent Two-Sample t-test was selected as the primary analytical method, since the study compared average Door-to-Doctor Time between two independent patient groups. Door-to-Doctor Time is a continuous numerical variable; the Control and Treatment groups consisted of different patients, and the sample of over two thousand encounters provided sufficient statistical power to detect meaningful differences.

Hypotheses

H₀ (Null): The AI-assisted triage workflow does not reduce the average Door-to-Doctor Time compared with the standard nurse-led process.

H₁ (Alternative): The AI-assisted triage workflow does reduce the average Door-to-Doctor Time compared with the standard nurse-led process.

Results

Metric	Result	Interpretation
Mean Difference	7.4 minutes	Patients assigned to AI-assisted triage reached a physician an average of 7.4 minutes earlier.
Performance Improvement	15.8%	A meaningful reduction in physician waiting time.
p-value	8.828×10^{-17}	Overwhelming statistical evidence against the null hypothesis.
95% CI	(-9.13, -5.67)	The true reduction in Door-to-Doctor Time is likely between 5.67 and 9.13 minutes.

Cohen's d	-0.353	A small-to-moderate practical effect.
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The p-value of 8.828×10^{-17} is far smaller than the conventional 0.05 threshold, meaning the probability of observing a reduction this large purely by chance is extraordinarily small. Because the entire 95% confidence interval falls below zero, the analysis consistently supports the conclusion that the AI-assisted workflow improves physician response times. The Cohen's d of -0.353 indicates a small-to-moderate effect size, modest in statistical terms, but meaningful operationally when applied across thousands of patient encounters.

4. Results & Business Impact

Patients assigned to the AI-assisted workflow reached a physician 7.4 minutes earlier on average, a 15.8% improvement over the standard nurse-led process. Importantly, this gain was achieved without introducing a meaningful bottleneck earlier in the journey: Door-to-Triage Time increased by only about seven seconds, an operationally negligible cost.

A reduction of 7.4 minutes may appear modest at first glance, but its implications compound across a busy Emergency Department that processes hundreds or even thousands of visits each month; small per-patient improvements accumulate into substantial department-wide gains. Earlier physician assessment allows diagnostic investigations, clinical decision-making, and treatment pathways to begin sooner, contributing to smoother patient flow, reduced congestion, and more efficient use of clinical resources and staff time.

Rather than suggesting AI replaces healthcare professionals, the findings reinforce that AI can function as an effective decision-support tool that complements clinical expertise while helping hospitals operate more efficiently.

5. Executive Dashboard

To translate statistical findings into a format suited for rapid interpretation by hospital leadership, an interactive executive dashboard was built in Microsoft Power BI. Designed for hospital executives and operational leaders, it presents six KPI cards summarizing average Door-to-Doctor Time, average Door-to-Triage Time, overall performance improvement, total patient encounters, and Control/Treatment group sizes.

Supporting charts show the patient distribution across groups (confirming near-equal randomization), a comparison of Door-to-Doctor Time highlighting the AI-assisted reduction, and a Door-to-Triage comparison showing the negligible increase in triage duration. The dashboard concludes with an executive recommendation panel, giving decision-makers a concise, evidence-based takeaway without needing to interpret each chart independently.

6. Executive Recommendation

Riverside General Hospital should consider proceeding with a phased implementation of the AI-assisted triage workflow rather than an immediate hospital-wide deployment. While the statistical evidence is strong, this evaluation used simulated data, so live implementation should occur gradually to allow continuous monitoring and refinement.

- Pilot Deployment — introduce the AI-assisted workflow within a single Emergency Department unit while monitoring operational performance, patient safety, and staff adoption.
- Performance Evaluation — compare pilot results against baseline operational metrics to confirm the improvements observed experimentally are replicated under real-world conditions.
- Scaled Implementation — expand deployment across additional departments only after consistent operational improvements are demonstrated, and clinical stakeholders express confidence in the workflow.

7. Limitations & Future Work

Limitations

- The analysis used a simulated dataset rather than operational hospital data, which cannot fully capture the complexity of real clinical operations.
- The evaluation focused on operational metrics only; clinical outcomes such as mortality, diagnostic accuracy, and adverse events were out of scope.
- The experiment covered a relatively short evaluation period, limiting insight into seasonal fluctuations and long-term adoption effects.
- The project did not assess healthcare professionals' perceptions of the workflow, though staff trust and usability are critical to successful implementation.

Future Improvements

- Evaluate AI-assisted triage in live hospital environments using real operational data, including multi-site studies across hospitals of different sizes.
- Incorporate additional performance measures, ED Length of Stay, patient satisfaction, Patients Leaving Without Being Seen, and readmission rates.
- Add economic evaluation and cost-benefit analysis to assess whether operational gains justify implementation and maintenance costs.
- Continue to assess fairness and algorithmic bias to ensure equitable performance across patient populations.

8. Skills Demonstrated

Technical

- Python · Pandas · NumPy · SciPy · Power BI · DAX · Plotly

Statistical

- Randomized A/B Testing · Hypothesis Testing · Confidence Interval Interpretation · Effect Size Analysis · Experimental Design

Healthcare Analytics

- Emergency Department Operations · KPI Development · Operational Performance Analysis · Healthcare Business Understanding

Business

- Executive Reporting · Business Problem Framing · Evidence-Based Decision Making · Strategic Recommendations

Communication

- Data Storytelling · Dashboard Design · Technical Writing · Executive Presentation

9. Professional Reflection

This project represents an important milestone in my development as a healthcare data scientist. At the outset, my focus was largely technical; success meant building dashboards, writing code, and producing statistically significant results. As the project progressed, my perspective evolved: successful data science begins long before statistical analysis, with understanding the business problem, selecting an appropriate methodology, and generating evidence that supports meaningful decision-making.

Designing a randomized A/B experiment challenged me to think beyond descriptive analytics and consider how evidence is generated in real-world organizational settings. Interpreting statistical results within a healthcare context reinforced the importance of translating technical findings into recommendations decision-makers can understand and act on.

The most valuable lesson: effective healthcare analytics is not simply about analyzing data; it is about improving decisions that ultimately influence patient care.

10. Conclusion

Through a randomized A/B testing framework, this project demonstrated that an AI-assisted triage workflow can meaningfully reduce Emergency Department Door-to-Doctor Time without introducing operationally significant delays during triage. The combination of exploratory analysis, statistical testing, and executive dashboard reporting provided a comprehensive evaluation of the intervention and translated analytical findings into actionable business recommendations.

As healthcare continues to adopt data-driven technologies, the ability to design rigorous experiments, interpret statistical evidence, and communicate findings effectively remains essential. This project reflects that philosophy, demonstrating how analytics can improve healthcare operations while maintaining a strong focus on patient-centered outcomes.